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# China Diesel Consumption Forecast 2026-2030

# 2026-2030 年中国柴油消费展望


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## Abstract 摘要

This report analyzes the trends of China's apparent diesel consumption from 2026 to 2030, adopting two perspectives: a bottom-up perspective based on heavy-duty trucks (HDT) and a top-down macro perspective.

In the first nine months of 2025, China's apparent diesel consumption has maintained a downward trend, with total annual consumption in 2025 projected at 185 million tons, mainly due to weakened road transportation demand. As the key diesel consumer in road transport, HDTs make up over 40% of China's total diesel use.

In 2026, the decline rate of China's apparent diesel consumption is likely to stabilize as 2025's HDT "trade-in" policy pre-empted 2026's replacement demand, and diesel HDT stock remains resilient amid contracted substitution by new energy (NEV) and LNG-powered HDTs. Total apparent diesel consumption for 2026 is estimated at 171-173 million tons.

Two forecasting methods (scrappage calculation and macroeconomic data-driven analysis) indicate that from 2026 to 2030, diesel HDT stock will be increasingly squeezed by NEVs and LNG HDTs (2030 sales penetration: 39% for NEVs, 30% for LNG HDTs). With the resulting decrease in HDT ownership volume, China's diesel consumption is projected to contract to 100-113 million tons by 2030.

**Risks:** Changes in the sales structure of HDTs; Macroeconomic fluctuations; Policy adjustments.

报告分别从自下而上的重卡视角和自上而下的宏观视角，分析了 2026-2030 年中国柴油消费趋势。

2025 年 1-9 月中国柴油表观消费同比延续下跌，全年预计达 1.85 亿吨。核心驱动因素为公路交通柴油消费下滑。而重卡柴油消费占国内柴油总消费的 40%，是公路柴油消费的核心支柱。

2026 年柴油表观消费降幅或持稳，主要因 2025 年重卡“以旧换新”政策提前透支 2026 年更新需求，新能源和 LNG 重卡可替代量收缩，柴油重卡存量显现韧性，预计 2026 年柴油消费为 1.71-1.73 亿吨。

2026-2030 年，通过重卡报废测算、宏观数据测算两种方法预测，若柴油重卡存量持续被新能源和 LNG 重卡挤压（2030 年新能源重卡销量渗透率预计达 39%，LNG 重卡达 30%），柴油重卡保有量将逐步下降，带动国内柴油消费向 1.0-1.13 亿吨区间收缩。

**风险因素：**重卡销量结构超预期变化，宏观波动，政策调整

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# Charts 图 表

# 1. China's Diesel Consumption Review

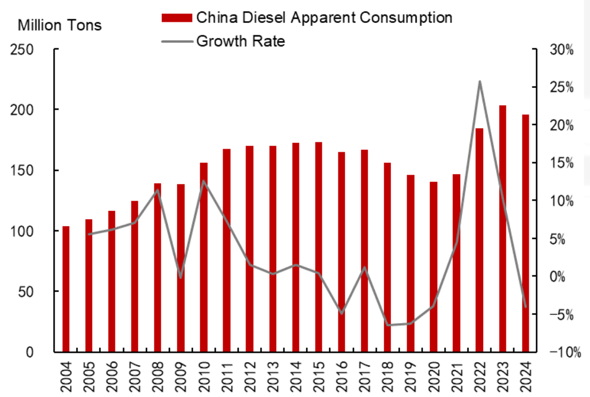
## 1.1 Peak in Total Volume, Highway Traffic Dominates

**China's diesel consumption has entered a downward cycle due to factors including economic restructuring and accelerated energy substitution.** The apparent consumption reached approximately 197 million tons in 2024, and is expected to be around 185 million tons in 2025, representing a year-on-year decrease of 6%.

**The transportation, storage, and postal services sector (hereinafter referred to as the "TSP sector") is a core pillar of China's diesel consumption, and has driven the overall consumption downward after hitting its peak.** From the perspective of consumption structure, the TSP sector has accounted for a stable 60-68% of total diesel consumption since 2012. In 2022, diesel consumption in the TSP sector reached approximately 158 million tons, accounting for 64%, making it the largest diesel-consuming segment. In recent years, the agriculture, forestry, animal husbandry, and fishery sector and the industrial sector have emerged as new growth drivers for diesel consumption, but their shares—11% and 9% respectively—are significantly lower than that of the TSP sector, making it difficult to reverse the downward trend of diesel consumption after the TSP sector's long-term peak in diesel use.

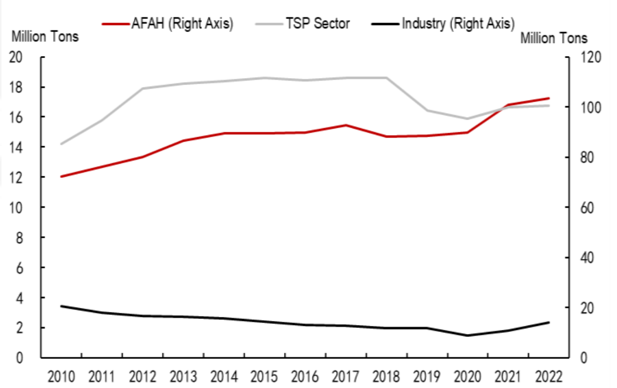
**Road transportation is the absolute mainstay of diesel consumption in the TSP sector and the primary battlefield for diesel substitution.** Over the past decade, road freight volume has accounted for more than 73% of China's total freight volume. Its core carrier, trucks, also features fast renewal cycles and high maturity of substitution technologies, solidifying road freight's position as the primary battlefield for diesel substitution.

**Chart 1: China Diesel Apparent Consumption**




Sources: Wind, CITIC Futures

**Chart 2: China Diesel Consumption by Sector**



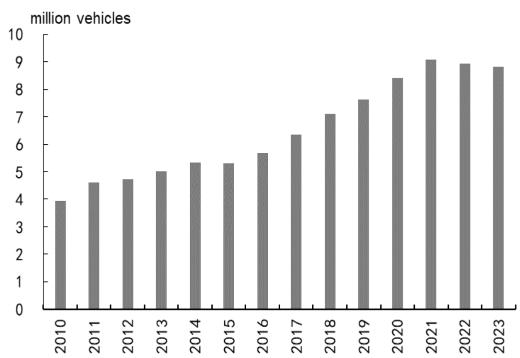

Sources: Wind, CITIC Futures

## 1.2 Peak in Diesel Heavy-Duty Truck Sales, Partial Substitution by NEV

**Diesel heavy-duty trucks (HDTs), characterized by high fuel consumption intensity and large operational scale, account for over 40% of China's total diesel consumption.** As the dominant model in road freight, their core position is jointly determined by their fuel consumption intensity and operational scale. Currently, more than 90% of cargo transportation in trunk-line logistics relies on diesel HDTs. With an average annual mileage exceeding 150,000 kilometers per vehicle and an average fuel consumption of 30-33 liters per 100 kilometers, diesel HDTs consume more than twice the fuel of medium-duty trucks and over four times that of light-duty trucks. From a market structure perspective, in 2024, medium and heavy-duty trucks accounted for over 65% of the diesel commercial vehicle market and contributed more than 75% of the total social road freight volume.

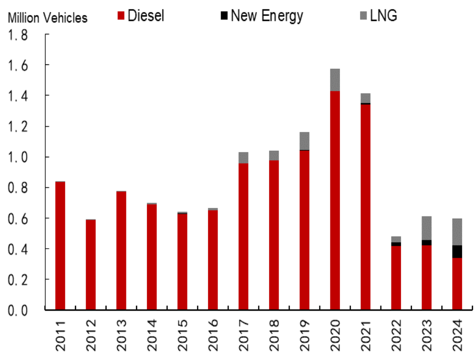
**However, diesel HDT ownership has been on a downward trend since 2021, with new energy HDTs accelerating their market penetration.** After reaching an all-time high of 1.43 million units in 2020 (based on compulsory traffic insurance data), domestic sales of diesel HDTs entered a downward trajectory, with a compound annual growth rate (CAGR) of -30.2% from 2021 to 2024. Furthermore, as new energy technologies mature and their economic viability improves, new energy HDTs have accelerated their market penetration. In recent years, driven by policy dividends and fuel cost advantages, LNG-powered HDTs achieved a high sales penetration rate of 30% in 2024. Currently, the industry has entered a phase of competition for existing market share. With the recovery of natural gas HDTs supported by favorable oil-gas price spreads, coupled with policy incentives for phasing out old diesel vehicles, the share of diesel HDTs in stock replacement has been continuously eroded. The growth momentum of diesel HDT ownership has completely shifted to contraction, and diesel vehicles will gradually retreat to trunk-line long-distance transportation and other areas that are difficult for new energy vehicles to cover.

**Chart 3: China Heavy Truck Ownership**



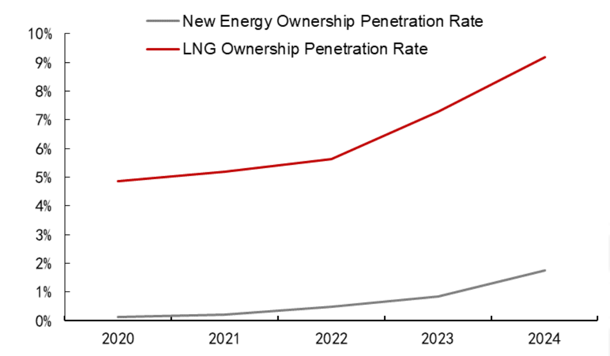

Sources: Wind, CITIC Futures

**Chart 4: China Heavy Truck Sales by Type**



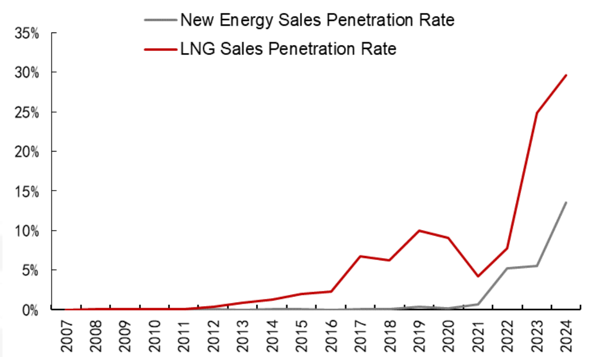

Sources: Wind, CITIC Futures

**Chart 5: China New Energy & LNG Heavy Truck Fleet Penetration Rate**




Sources: Wind CITIC Futures

**Chart 6: China New Energy & LNG Heavy Truck Sales Penetration Rate**



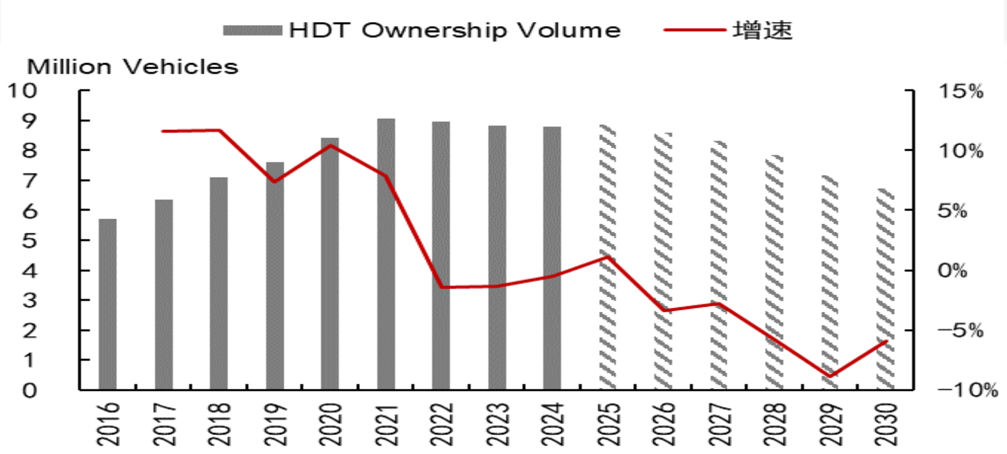

Sources: Wind, CITIC Futures

# 2. 2026-2030 Heavy-Duty Truck Ownership Volume Forecast

## 2.1 Method 1: Scrappage Calculation

Based on the assumptions of the natural scrappage period and average scrappage age, we forecast **China's HDT ownership volume for 2025-2030 as 8.87/8.57/8.33/7.85/7.15/6.73 million units**, and sales volume as 0.76/0.73/0.80/0.68/0.87/0.99 million units respectively. The ownership volume will be gradually shrinking, while HDT sales will follow a "decline-then-recovery" trajectory, with the inflection point occurring in 2029.

**Chart 7: Method 1, 2016-2030 China's HDT Ownership Volume Forecast**




Sources: NBS, CMI, CITIC Futures

**The theoretical natural scrappage period for HDTs is 9 years, and natural scrappage volume is expected to surge significantly during 2025-2030.** We assume the natural scrappage period is T years. Meanwhile, the average HDT sales volume over the range of [T-1, T+1] years is used as the theoretical scrappage volume for the current year. The actual scrappage volume in a given year is calculated as "current-year sales volume - change in ownership volume." We then compute the ratio of theoretical scrappage volume to actual scrappage volume. The closer the ratio is to 100%, the more accurate the estimation of the scrappage period. Ultimately, we find that when the HDT scrappage period is set at 9 years, the theoretical and actual scrappage volumes are most aligned. Therefore, the natural scrappage volume for 2025-2030 is determined based on domestic sales volume nine years prior (i.e., scrappage volume for 2025-2030 corresponds to the sales boom period of 2016-2021), indicating a substantial increase in natural scrappage volume during 2025-2030.

**The actual average scrappage age of HDTs is expected to remain at a high level initially and then gradually decline.** When the HDT industry features competition for existing market share, with limited new demand and growth driven mainly by replacement demand, the cycle enters a downward phase. Meanwhile, oversupply (more trucks than cargo), low freight rates, and poor profitability lead to extended scrappage periods. Consequently, scrappage periods are shorter during upward cycles, longer during downward cycles, and significantly deviate from the average at the bottom of the cycle. Currently, China's HDT industry is in a downward cycle,. Therefore, we anticipate the average scrappage age will remain 10-12 years during 2025-2028. However, as natural scrappage volume for 2029-2030 corresponds to the 2020-2021 sales boom, the HDT market structure may reach an inflection point, and the average scrappage age will gradually decrease to 7 years.

**The HDT scrappage rate will remain at a high level, increasing initially and then stabilizing.** Under the constraints of the aforementioned assumptions regarding the natural scrappage period and average scrappage age, we use three widely accepted industry formulas:

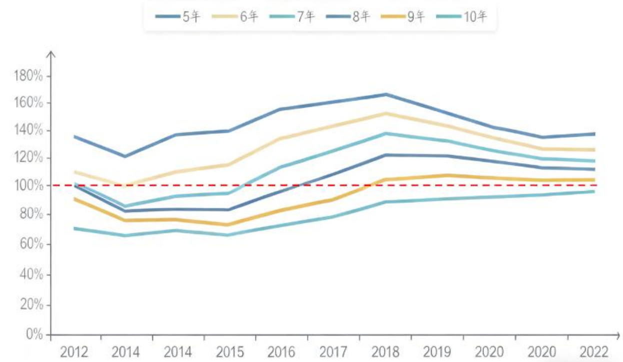
Ownership Volume = Scrappage Volume / Scrappage Rate

Sales Volume = Current-Year Ownership Volume - Previous-Year Ownership Volume + Scrappage Volume

Average Scrappage Age = Ownership Volume / Sales Volume

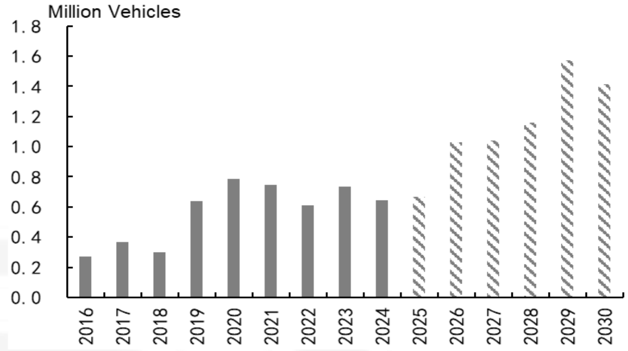
Following the logical chain of "current-year scrappage volume & scrappage rate → current-year HDT ownership volume → current-year sales volume → average scrappage age," the scrappage rate is essentially locked in with limited room for adjustment, as it is constrained by scrappage volume and average scrappage age at both ends of the calculation. The scrappage rate will increase from 7.5% in 2025 to 22% in 2029, and stabilize at 21% in 2030.

**Chart 8: Ratio of Theoretical to Actual Scrappage Volume of China's HDTs**



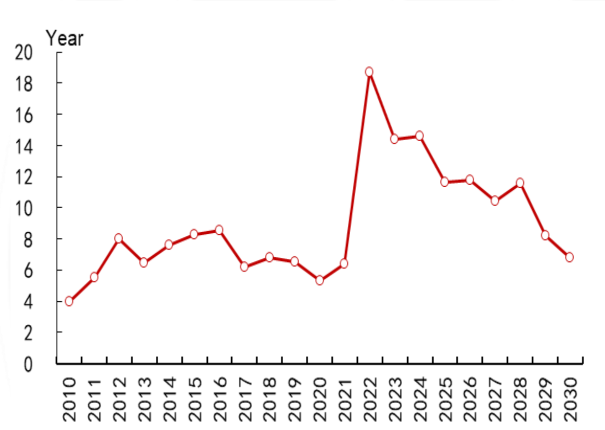

Sources: NBS, CMI, CITIC Futures

**Chart 9: 2016-2030 China's HDT Scrappage Volume**



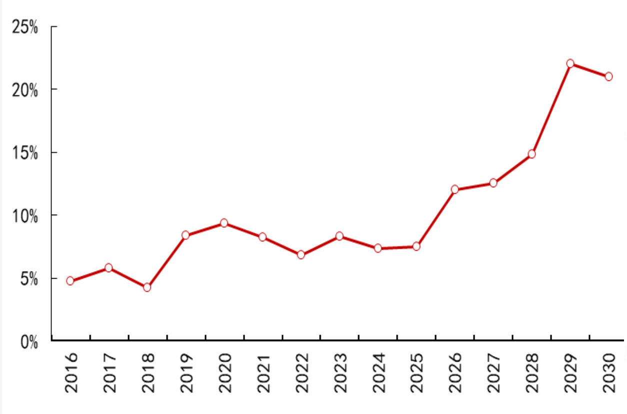

Sources: NBS, CMI, CITIC Futures

**Chart 10: 2010-2030 Average Scrappage Age of China's HDTs**




Sources: NBS, CMI, CITIC Futures

**Chart 11: 2021-2030 China's HDT Scrappage Rate**

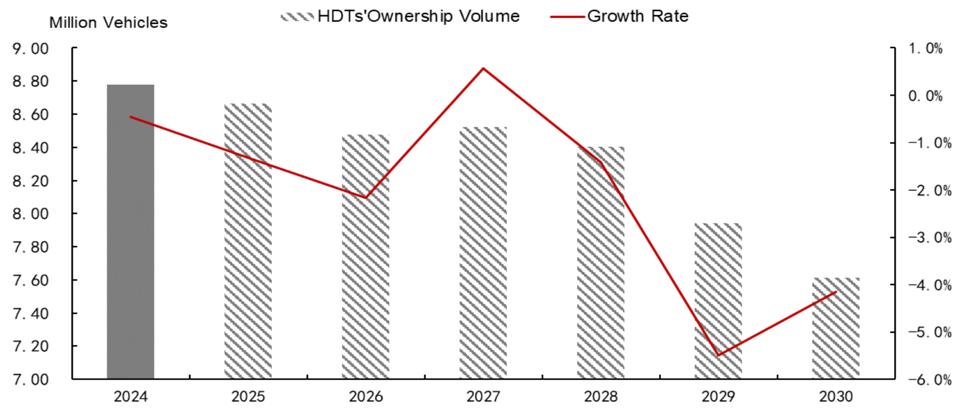



Sources: NBS, CMI, CITIC Futures

## 2.2 Method 2: Macroeconomic Data-Driven Analysis

Under the expectation of gradual macroeconomic recovery, we forecast total HDT ownership volume (logistics + engineering HDTs) for 2025-2030 at 8.77/8.47/8.52/8.40/7.94/7.61 million units, with total sales volume at 720,000/740,000/770,000/780,000/790,000/810,000 units.

**Chart 12: Method 2: 2025-2030 China's HDT Ownership Volume Forecast**




Sources: NBS, CMI, CITIC Futures

**HDTs can be categorized into logistics HDTs and engineering HDTs. In recent years, China's HDT market has been transitioning from an "engineering-dominated" to a "logistics-dominated" structure.** Logistics HDTs (semi-trailer tractors and complete HDT vehicles per CAAM's classification) account for approximately 60% of total industry demand. Primarily used for long-distance inter-provincial cargo transportation, their demand is correlated with manufacturing sentiment and macro liquidity. Engineering HDTs (HDT chassis per CAAM's classification) represent around 34% of demand, including dump trucks, cranes modified from HDT chassis, and concrete machinery. These are mainly utilized in real estate construction and infrastructure projects.

**Manufacturing and industrial sectors are expected to recover moderately year-on-year, driving steady growth in logistics HDT sales.** Considering that logistics HDT demand is influenced by manufacturing sentiment, industrial production, and macro liquidity, we use the Manufacturing PMI and PMI New Orders Index to represent manufacturing sentiment, power generation to reflect actual industrial production, and M2 (broad money supply) plus import-export volume to indicate macro liquidity structure and cross-border logistics demand. Our projections for 2026-2030 are as follows:

- Manufacturing PMI is expected to remain above the boom-bust line, supported by recovering industrial production and steady growth in logistics demand; the PMI New Orders Index is projected to grow faster than the overall PMI, driven by rebounding export demand.
- M2 will increase steadily under a moderately accommodative monetary policy.
- Import-export volume is likely to rise amid infrastructure demand along the "Belt and Road" initiative.
- Power generation is expected to grow steadily with industrial recovery and new energy infrastructure development.

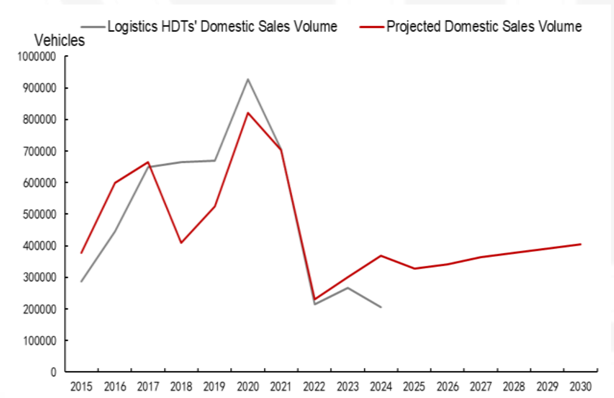
Based on the fitting results of these indicators, we forecast **domestic sales of logistics HDTs for 2025-2030 as 328,000/342,000/365,000/378,000/392,000/405,000 units respectively. Adopting the same scrappage period assumptions as above, the ownership volume of logistics HDTs is projected to reach 5.58/5.26/5.21/5.06/4.64/4.34 million units.**

**The real estate sector has not yet reached an inflection point, while industrial development will drive modest growth in engineering HDT sales.** In terms of demand structure, approximately 80% of engineering HDT demand is concentrated in the real estate sector, making real estate development investment the core indicator for projections, with a correlation coefficient of 0.81 between real estate development investment and domestic engineering HDT sales. Benefiting from stock improvement in residential investment, the development of industrial real estate, and policy dividends from urban renewal, we forecast real estate investment volume for 2026-2030 at 148/151/153/155/158 trillion yuan.

**Moreover, total industrial value-added is used as a correction indicator to cover "engineering demand outside the real estate sector”.** Driven by medium-speed growth in high-tech manufacturing, green manufacturing, and digital transformation, total industrial value-added is expected to reach 450/470/491/513/530 trillion yuan during 2026-2030.

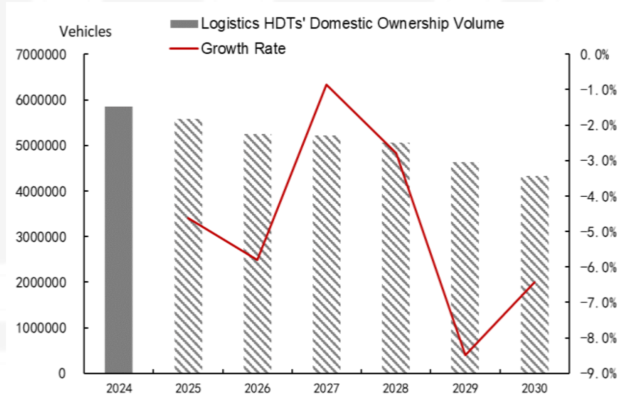
Based on the linear equation fitted from these indicators, domestic sales of engineering HDTs for 2025-2030 are projected to be 388,000/396,000/403,000/403,000/403,000/408,000 units, **with ownership volume at 2.93/3.08/3.22/3.31/3.34/3.27 million units respectively.**

**Chart 13: 2015-2030 Logistics HDTs' Sales Volume**



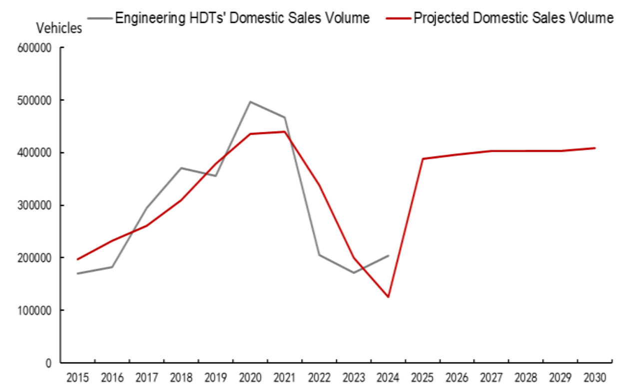

Sources: Wind, CITIC Futures

**Chart 14: 2024-2030 Logistics HDTs' Domestic Ownership Volume**



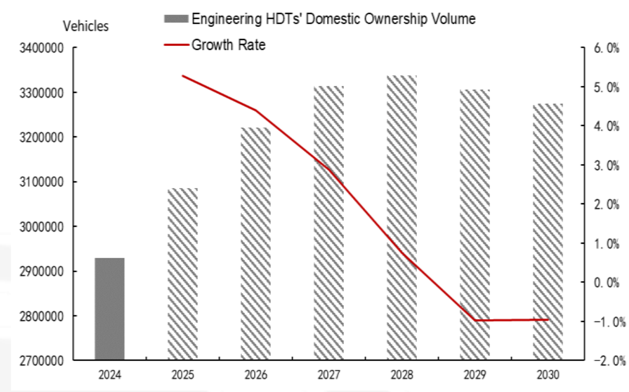

Sources: Politico, CITIC Futures

**Chart 15: 2015-2030 Engineering HDTs' Domestic Sales Volume**




Sources: Wind, CITIC Futures

**Chart 16: 2024-2030 Engineering HDTs' Ownership Volume**




Sources: Wind, CITIC Futures

## 2.3 Comparison of the Two Methods

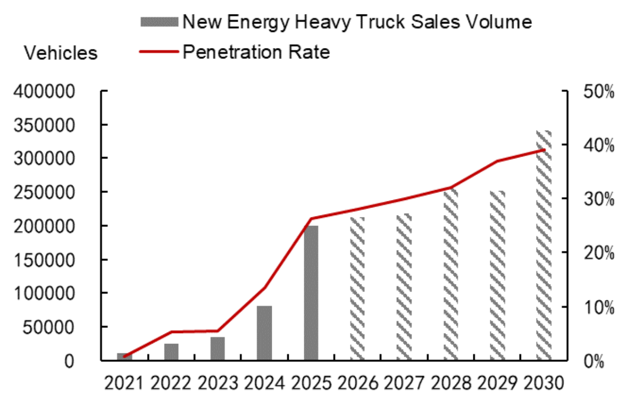
**Driven by the accelerated penetration of new energy and LNG-powered heavy-duty trucks (HDTs), the share of diesel HDT ownership volume is expected to gradually decline to 50%.** In terms of the pace, currently constrained by technical bottlenecks in driving range and energy replenishment efficiency of electric HDTs, as well as the limited number of charging piles, both the ownership and sales volumes of electric HDTs are lower than those of LNG-powered HDTs. However, as LNG serves as a "transitional clean energy" for diesel vehicles, the dual carbon goals, coupled with technological innovations in electric HDTs and the commercialization of solid-state battery technology, will lead to long-term squeeze on LNG-powered HDTs by new energy alternatives. The growth rate of LNG-powered HDTs' penetration rate will be lower than that of new energy HDTs. Therefore, we assume that by 2030, the sales penetration rate of new energy HDTs will reach 39%, and that of LNG-powered HDTs will reach 30%, with both increasing year by year and accelerating between 2029-2030. Notably, the sales penetration rate of new energy HDTs will grow faster than that of LNG-powered HDTs. By 2030, the sales share of diesel HDTs will drop to 31%, and their ownership volume will decline to 50%.

Comparing the Two Methods, the projected diesel HDT ownership volumes for 2025-2030 are as follows:

**Methodology 1: 7.53/6.91/6.28/5.51/4.38/3.34 million units;**

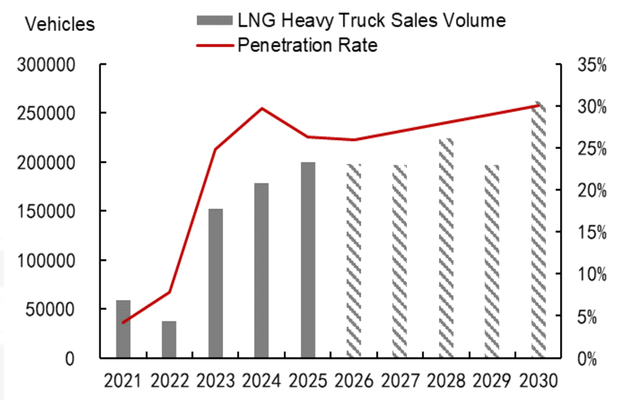
**Methodology 2: 7.36/6.83/6.42/5.90/4.86/3.78 million units.**

**Chart 17: 2021-2030 New Energy Heavy Truck Sales Penetration Rate**



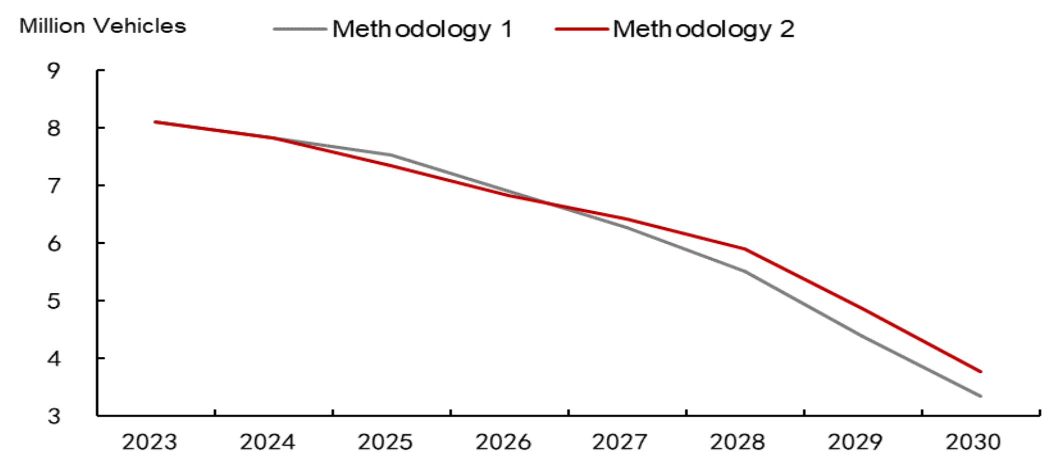

Sources: NBS, CMI, CITIC Futures

**Chart 18: 2021-2030 LNG Heavy Truck Sales Penetration Rate**




Sources: NBS, CMI, CITIC Futures

**Chart 19: 2025-2030 Forecast of Diesel HDT Ownership Volume Under the Two Methods**




Sources: NBS, CMI, CITIC Futures

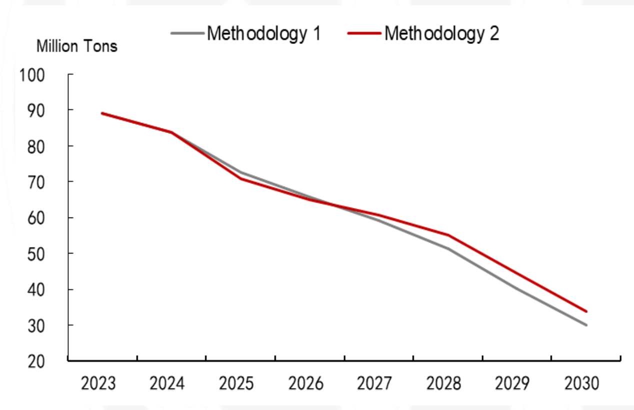
## 3. 2026-2030 China's Diesel Consumption Outlook

**Fuel consumption per diesel truck continues to decline overall.** In 2024, China had approximately 7.82 million diesel heavy trucks, which accounted for 43% of the country's total diesel consumption. China's apparent diesel consumption was about 190 million tons, with the average fuel consumption of diesel heavy trucks reaching about 10.7 tons per vehicle per year. With strengthened policies and technological improvements, the new GB 30510-2024 standard was implemented in July 2025, reducing fuel consumption limits by approximately 10%. We assume that from 2025 to 2028, fuel consumption per conventional diesel truck will decline by 1% annually, and by 2% annually from 2029 to 2030, resulting in fuel consumption of 9.6/9.5/9.4/9.3/9.2/9.0 tons per vehicle per year respectively.

**Diesel consumption by diesel heavy trucks = diesel heavy truck fleet × annual fuel consumption per vehicle.** According to Method 1, diesel consumption by diesel heavy trucks is calculated to be 72.55/65.88/59.26/51.48/40.09/29.99 million tons; according to Method 2, it is calculated to be 70.84/65.14/60.63/55.11/44.53/33.94 million tons.

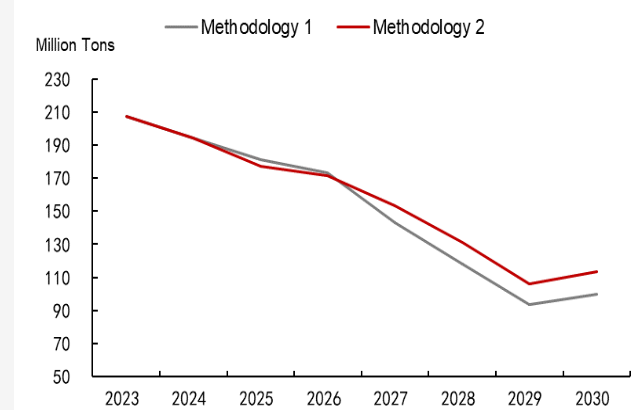
**The proportion of diesel consumed by heavy trucks in China's total apparent diesel consumption decreases year by year.** With the accelerated replacement of diesel heavy trucks, their share in total diesel consumption gradually shrinks. We assume that from 2026 to 2030, this proportion will be 38%/36%/34%/32%/30% respectively. **Therefore, according to Method 1, China's apparent diesel consumption would be 173/143/118/94/100 million tons; according to Method 2, it would be 171/153/131/106/113 million tons.**

**Chart 20: 2026-2030 Forecast of Diesel Consumption by Diesel HDTs Under the Two Methods**




Sources: NBS, CMI, CITIC Futures

**Chart 21: 2026-2030 Forecast of China's Apparent Diesel Consumption Under the Two Methods**




Sources: NBS, CMI, CITIC Futures

# Appendix: Chinese Version

## 1. 中国柴油需求回顾

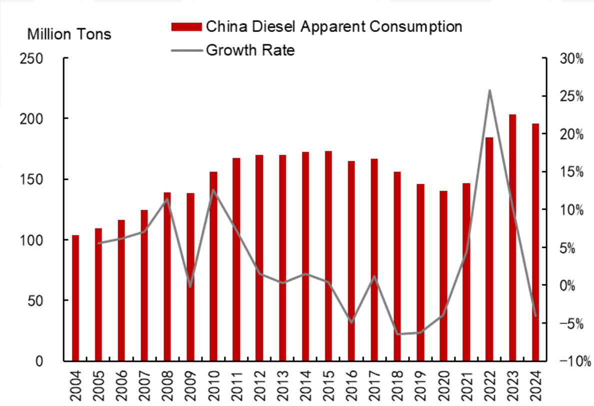
### 1.1 总量达峰，公路交通主导

**中国柴油消费因经济结构转型、能源替代加速等进入下行周期。**2024 年表表观消费量约 1.97 亿吨，2025 年预计约 1.85 亿吨，同比-6%。

**交仓邮业是中国柴油消费的核心支柱，达峰后带动总消费下行。**结构来看，交通运输、仓储和邮政业（以下简称“交仓邮业”）的消费量占比自 2012 年以来基本稳定在 60-68%，2022 年交仓邮业柴油消费约 1.58 亿吨，占比 64%，始终是柴油消费的核心领域。近几年，农林牧渔业和工业成为柴油消费的新增长点，但是占比显著低于交仓邮业，分别为 11%、9%，难以扭转柴油消费颓势。

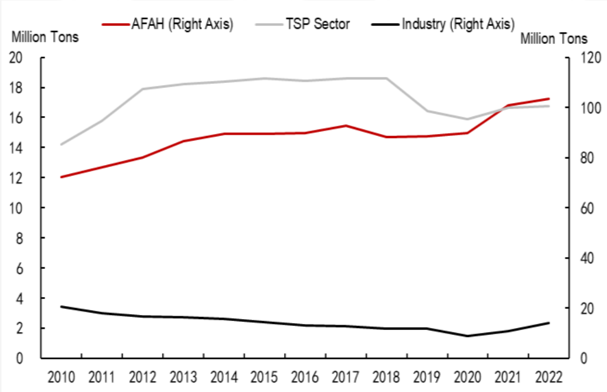
**公路运输是交仓邮业柴油消费的绝对主力，也是柴油替代的主战场。**过去 10 年，公路货运量均占国内总货运量的 73%以上。公路运输的其核心载体——载货汽车，更新迭代速度快、替代技术成熟度高，成为柴油替代的主战场。

图表1：中国柴油表观消费量与增速




Sources: Wind, CITIC Futures

图表2：中国柴油消费-分部门



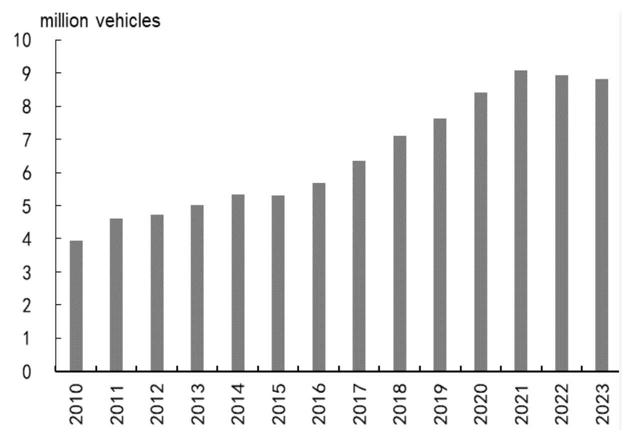

Sources: Wind, CITIC Futures

### 1.2 柴油重卡销量达峰，新能源部分替代

**柴油重卡油耗强度和运营规模双高，占国内柴油总消费的 40%以上。**柴油重卡是公路货运的主力车型，关注其油耗强度与运营规模。90%以上的干线物流货物运输依赖柴油重卡完成，单车年均行驶里程超 15 万公里，平均油耗达 30-33L/100km，是中型货车的 2 倍、轻型货车的 4 倍以上。2024 年柴油商用车市场中，中重型货车占比超 65%，货运量占比全社会公路货运总量 75%以上。

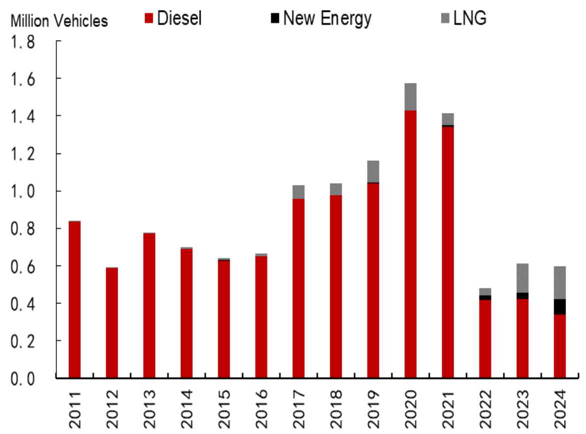
**2021年柴油重卡保有量进入下滑通道,新能源加速主导短途运输市场。**我国柴油重卡国内销量在2020年达到巅峰143万辆(交强险口径)后进入下行通道,2021-2024年柴油重卡销量同比年均增速为-30.2%。然而，随着新能源技术逐渐成熟和经济性提高，新能源正在加速主导短途运输市场。近年来，在政策红利和燃料成本优势驱动下，LNG重卡在2024年销量渗透率高达30%。当前行业已进入存量竞争阶段，天然气重卡因油气价差优势复苏，叠加政策对老旧柴油车的淘汰激励，柴油重卡在存量替换中的份额持续被挤压。柴油重卡保有量增长动能萎缩，或将逐步退守长途干线等新能源暂时难以覆盖的领域。

**图表3：中国重卡总保有量**



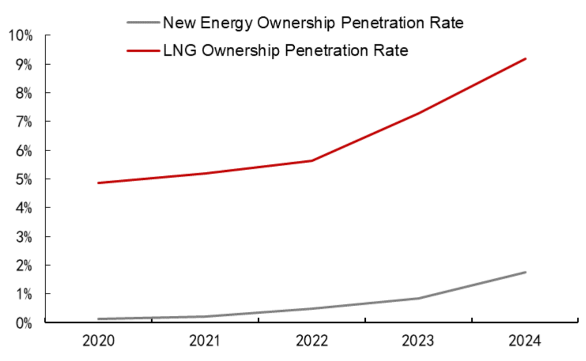

Sources: Wind, CITIC Futures

**图表4：中国重卡分车型销量**



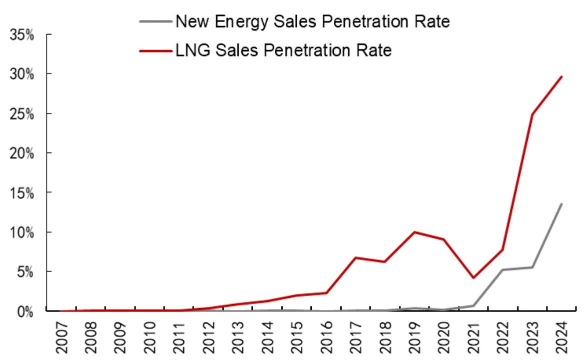

Sources: Wind, CITIC Futures

**图表5：中国新能源与LNG重卡保有量渗透率**




Sources: Wind CITIC Futures

**图表6：中国新能源与LNG重卡销量渗透率**



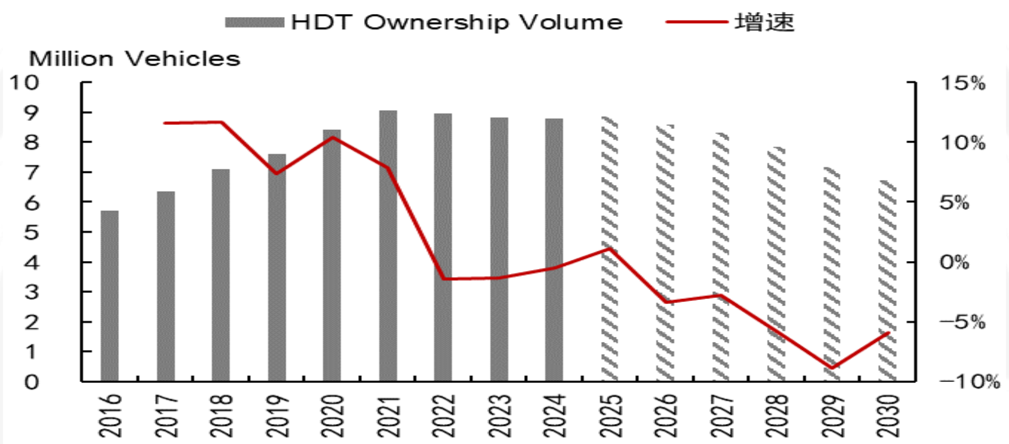

Sources: Wind, CITIC Futures

# 2. 2026-2030 年重卡保有量预测

## 2.1 方法一：重卡报废测算

**在自然报废年限和平均报废年限的假设下，预计 2026-2030 年国内重卡保有量分别为 857/833/785/715/673 万辆，重卡销量分别为 73/80/68/87/99 万辆。**重卡保有量逐渐萎缩，重卡销量先降后升，拐点出现于 2029 年。

**图表7：方法一：2016-2030 年国内重卡保有量预测**




Sources: NBS, CMI, CITIC Futures

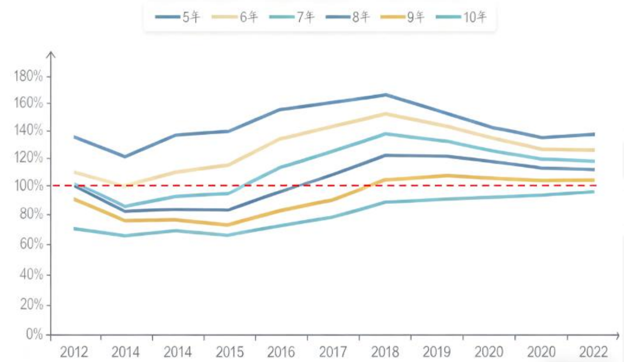
**测算估计重卡理论自然报废年限为 9 年，2025-2030 年自然报废量或将大幅增长。**假设自然报废年限为 T 年，同时以【T-1，T+1】年的重卡销量均值作为当年理论报废量，当年的实际报废量则为“当年销量-保有量变化量”，计算理论报废量/实际报废量的比值。若该比值越接近 100%，则对报废年限的估算越接近真实值。最终发现重卡报废年限为 9 年时，理论报废量和实际报废量水平最为接近。因此，2025-2030 年报废量对应 2016-2021 年的销量爆发期，预计 2025-2030 年的自然报废量大幅增加。

**重卡实际平均报废年限预计先维持高位水平，后逐渐下降。**上行期报废年限低，下行期报废年限长，周期底部报废年限大幅偏离中枢。当重卡行业进入存量竞争格局，新增需求有限，主要依赖更新需求时，进入下行期。同时，货少车多、运价低迷导致盈利状况较差，报废年限拉长。当前，国内重卡行业已处于下行期，我们认为 2025-2028 年仍维持 10-12 年的平均报废年限；2029-2030 年的自然报废量对应 2020-2021 年的销量爆发期，重卡市场格局或迎来拐点，平均报废年限逐年下降至 7 年。

**重卡报废率维持在高水平，节奏上先增后稳。**在前面自然报废年限和平均报废年限的假设约束下，利用行业内普遍使用的衡量公式保有量=报废量/报废率”、“销量=当年保有量-去年保有量+报废量”、“平均报废年限=保有量/销量”的公式，通过“当年报废量和报废率—>当年重卡保有量—>当年销量—>平均报废年限”的逻辑链条，由于头尾由报废量和平均报废年限限制，测算发现报废率可以基本锁定，可调节

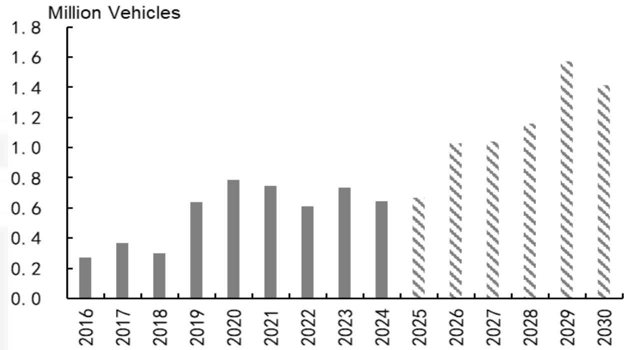
空间弹性较小。2025-2030 年报废率先增后稳，由 2025 年的 7.5%上涨至 2029 年的 22%，2030 年报废率为 21%。

**图表8：中国重卡理论报废量/实际报废量的比值**



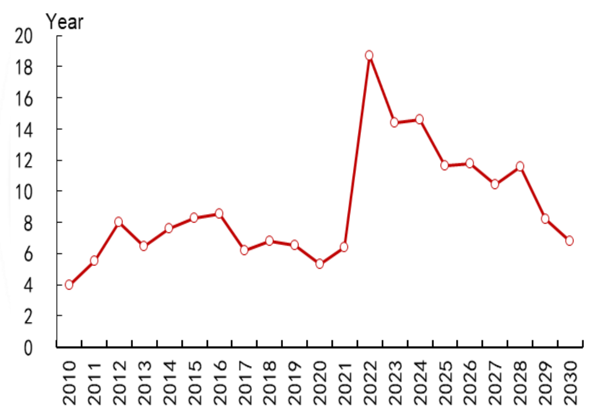

Sources: NBS, CMI, CITIC Futures

**图表9：2016-2030 年中国重卡报废量**



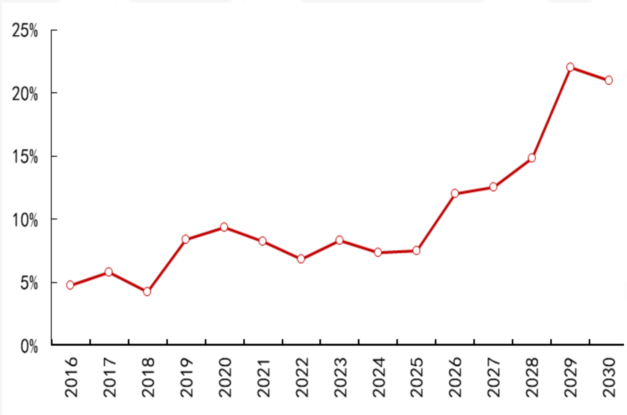

Sources: NBS, CMI, CITIC Futures

**图表10：2010-2030 年中国重卡平均报废年限**




Sources: NBS, CMI, CITIC Futures

**图表11：2021-2030 年中国重卡报废率**

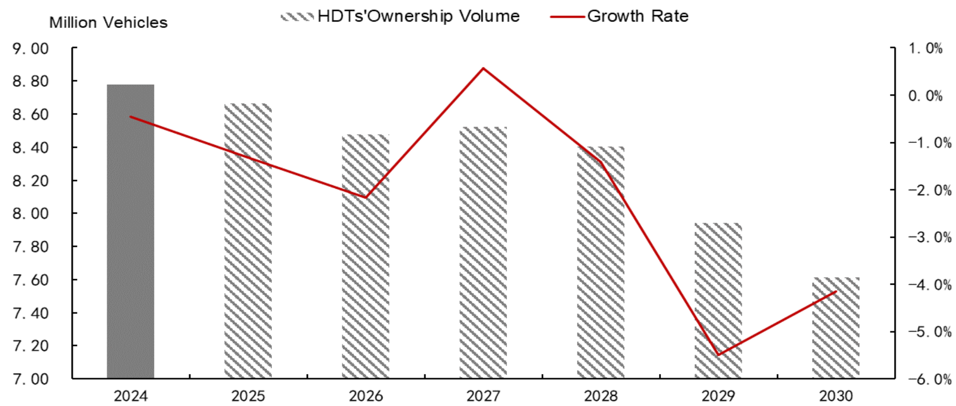



Sources: NBS, CMI, CITIC Futures

## 2.2 方法二：宏观数据测算

在宏观逐渐回暖的预期下，预计 2025-2030 年重卡保有量（物流重卡保有量+工程重卡保有量）为 877/847/852/840/794/761 万辆，销量预计为 72/74/77/78/79/81 万辆。

**图表12：方法二：2025-2030 年国内重卡保有量预测**



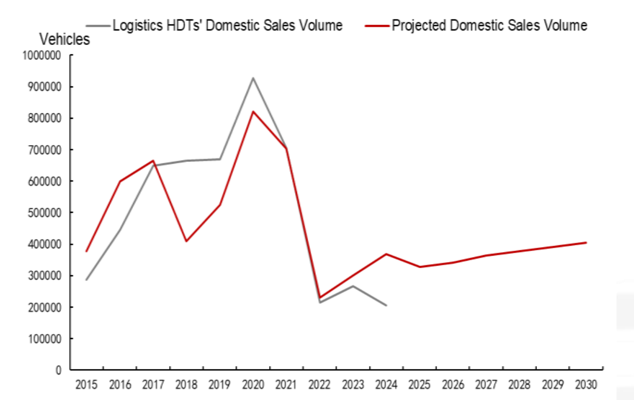

Sources: NBS, CMI, CITIC Futures

**重卡可分为物流重卡和工程重卡，从“工程主导”向 “物流主导”转型。**物流类重卡（中汽协口径的半挂牵引车和重卡整车）约占行业总需求的 60%，主要用于国内省际的长距离货物运输，其需求与制造业景气度和宏观流动性相关。工程类重卡（中汽协口径的重卡底盘）约占 34%，包括工程自卸车、重卡底盘改装的起重机、混凝土机械等，主要用于房地产建设和基建活动。

**物流重卡方面，预计制造业和工业预期逐年温和回暖，物流重卡销量逐年增加。**物流重卡受制造业景气度、工业生产和宏观流动性的影响，使用制造业 PMI 与 PMI 新订单指数代表制造业景气度，发电量代表工业生产的实际情况，M2 和进出口金额表示宏观流动性结构和跨境物流需求。我们认为，2026-2030 年制造业在工业生产回升和物流需求稳步增长的预期下，预计维持在荣枯线以上，PMI 新订单出口需求修复预期下预计整体增速高于整体 PMI；在适度宽松货币政策下，M2 稳步增长；在“一带一路”沿线基建需求下，进出口金额有望抬升；发电量在工业复苏和新能源基建预期下，有望稳步增长。因此，**根据各指标拟合的结果，预计 2025-2030 年物流重卡国内销量分别为 32.8/34.2/36.5/37.8/39.2/40.5 万辆，报废年限与前面的假设一致，保有量预计为 558/526/521/506/464/434 万辆。**

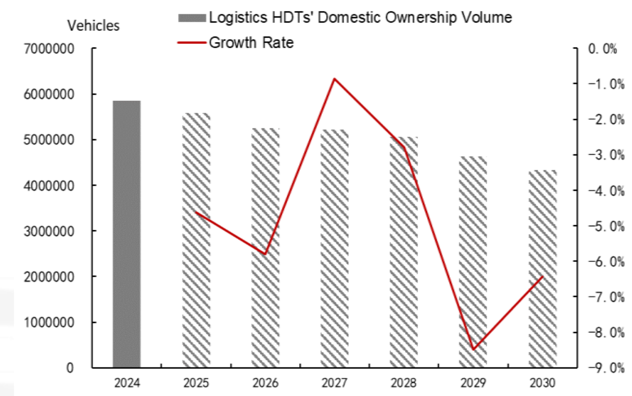
工程重卡方面，一是，地产行业仍未现拐点，工业发展提振工程重卡销量稳中小增。从需求结构来看，工程重卡 80%左右的需求集中于房地产领域，房地产开发投资与工程重卡国内销量相关系数高达 0.81。出于住宅投资的存量改善、产业地产的发展和城市更新的政策红利，预计 2026-2030 年房地产投资额为 14.8/15.1/15.3/15.5/15.8 万亿元。二是，工业全部增加值作为修正指标，覆盖“非房地产领域的工程需求”。2026-2030 年，高技术制造业、绿色制造与数字化转型带来的中速增长，工业全部增加值预计为 45.0/47.0/49.1/51.3/53.万亿元。**根据指标拟合的线性方程计算，预计 2025-2030 年工程重卡国内销量为 38.8/39.6/40.3/40.3/40.3/40.8 万辆，保有量分别为 293/308/322/331/334/330/327 万辆。**

**图表13：2015-2030 年物流重卡销量**



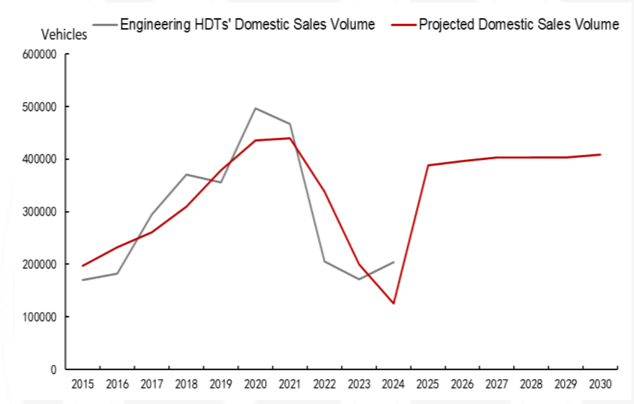

Sources: Wind, CITIC Futures

**图表14：2024-2030 年物流重卡保有量**



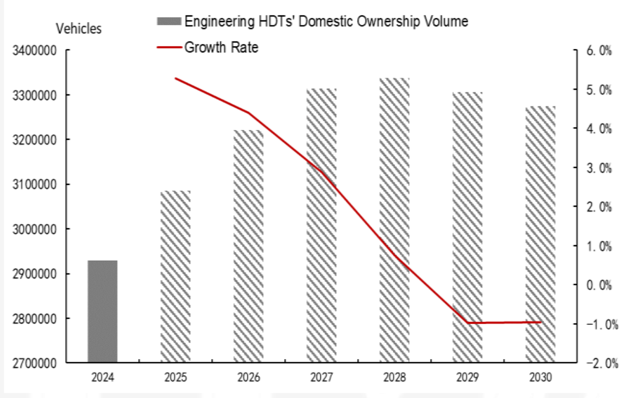

Sources: Politico, CITIC Futures

**图表15：2015-2030 年工程重卡销量**




Sources: Wind, CITIC Futures

**图表16：2024-2030 年工程重卡保有量**



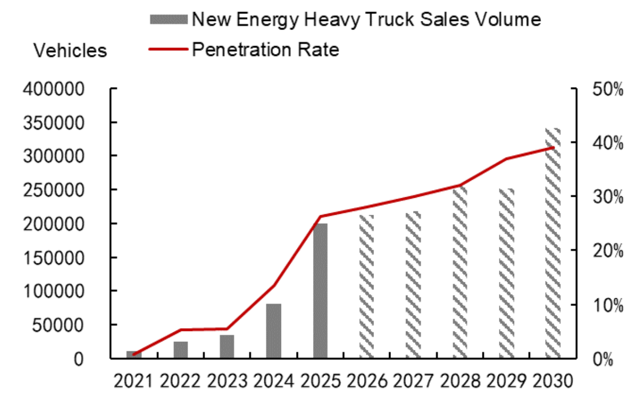

Sources: Wind, CITIC Futures

## 2.3 测算数据对比

**受新能源和 LNG 重卡渗透影响，柴油重卡保有量预计逐步下降至 50%。**节奏方面，当前电动重卡的续航与补能面临技术瓶颈，叠加充电桩保有量受限，电动重卡的保有量和销量都较 LNG 重卡低。但由于 LNG 作为柴油车的“过渡性清洁能源”，“双碳”目标叠加电动重卡技术革新和固态电池技术实现商业化应用，LNG 重卡长期会受新能源挤压，其渗透率增速或低于新能源。因此，假设 2030 年新能源重卡销量渗透率达到 39%，LNG 重卡销量渗透率达到 30%，两者逐年增加，2029-2030 年增速加快，且新能源销量渗透率快于 LNG。预计 2030 年，柴油重卡销量将降至 31%，柴油重卡保有量退坡至 50%。

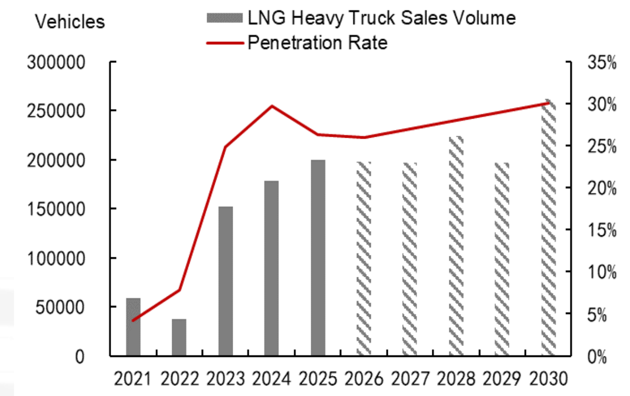
**对比两种测算方法，2025-2030 年，方法一中测算出柴油重卡保有量分别为 753/691/628/551/438/334 万辆；方法二中，测算出柴油重卡保有量分别为 736/683/642/590/486/378 万辆。**

图表17：2021-2030 年新能源重卡销量渗透率



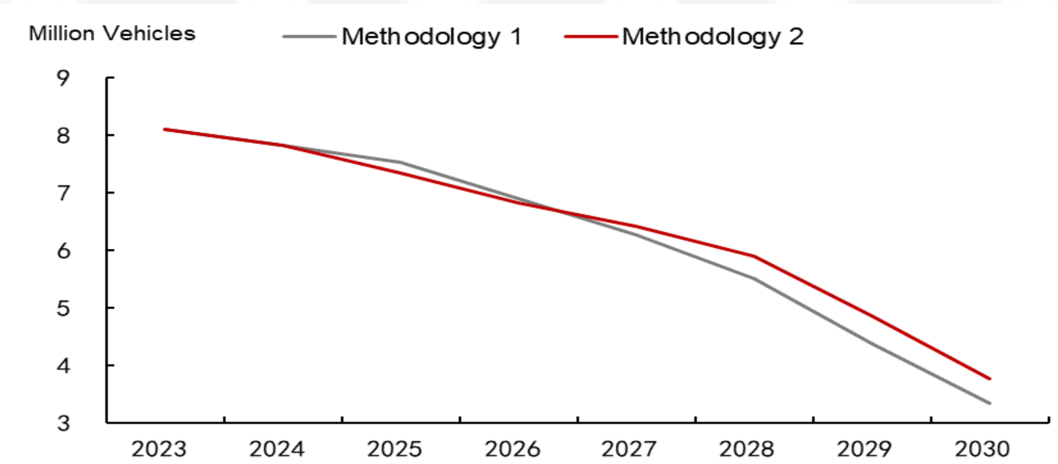

Sources: NBS, CMI, CITIC Futures

图表18：2021-2030 年 LNG 重卡销量渗透率




Sources: NBS, CMI, CITIC Futures

图表19：2025-2030 年两种测算方法下的柴油重卡保有量预测




Sources: NBS, CMI, CITIC Futures

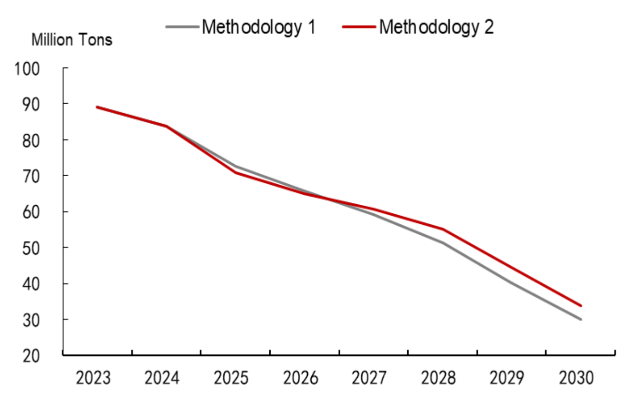
## 3. 2026-2030 年中国柴油消费展望

**单车油耗总体延续下滑趋势。**2024 年，柴油重卡保有量约 782 万辆，柴油重卡消费柴油占比国内柴油消费 43%，国内柴油表观消费量约 1.9 亿吨，柴油重卡平均油耗约 10.7 吨/年·辆。随着政策加码和技术提升，GB 30510-2024 新国标于 2025 年 7 月实施，油耗限值整体降低 10%左右。假设 2025-2028 年普通柴油重卡油耗平均下滑 1%，2029-2030 年下滑 2%，则分别为 9.6/9.5/9.4/9.3/9.2/9.0 吨/年·辆。

**柴油重卡柴油消费=柴油重卡保有量*单车年柴油消费。**在方法一中，测算得出柴油重卡消费柴油 7255/6588/5926/5148/4009/2999 万吨；在方法二中，测算得出柴油重卡消费柴油 7084/6514/6063/5511/4453/3394 万吨。

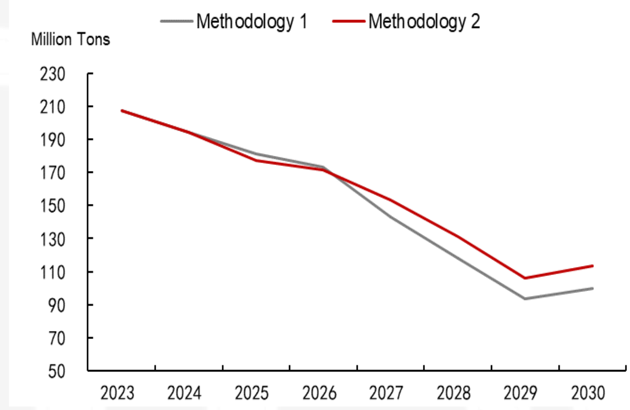
**柴油重卡占国内柴油表观消费量比例逐年下降。**随着柴油重卡替代加速，柴油重卡的柴油消费占比国内柴油消费占比逐渐缩小，假设 2026-2030 年占比分别为 38%/36%/34%/32%/30%。因此，在方法一中，国内柴油表观消费量分别为 1.73/1.43/1.18/0.94/1.0 亿吨；在方法二中，国内柴油表观消费量分别为 1.71/1.53/1.31/1.06/1.13 亿吨。

**图表20：2026-2030 年两种测算方式下的柴油重卡消费柴油量预测**




Sources: NBS, CMI, CITIC Futures

**图表21：2026-2030 年两种测算方式下的国内柴油表观消费量预测**




Sources: NBS, CMI, CITIC Futures

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